

Math Camp 2025 – Problem Set 6

Read the following problems carefully and justify all your work. Avoid using calculators or computers.

1. Matrix Arithmetic. Consider the following vectors and matrices:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 3 & 2 & -4 \\ -8 & 0 & 6 \end{pmatrix} \quad D = \begin{pmatrix} 6 & -2 \\ -1 & 3 \\ -3 & 8 \end{pmatrix} \quad a = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix} \quad b = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$$

Complete the following operations or give a reason why you cannot:

1. $3a - 2b$

8. $ab^\top$

2. $\|b\|$

9. $ab^\top B - DC$

3. $\|a - b\|$

10. bD

4. $\|Ca\|$

11. $A^\top A$

5. CD

12. $b^\top D$

6. DC

7. $a \cdot b$

13. B^2

Optional. What is a general formula for $B^n = \underbrace{B \cdot \dots \cdot B}_{n \text{ times}}$?